

Template

Studentnames and studentnumbers here

2026-06-17

Set-up your environment

```
require(tidyverse)
```

```
## Loading required package: tidyverse
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.3      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
require(rmarkdown)
```

```
## Loading required package: rmarkdown
```

```
require(ggplot2)
```

```
require(tinytex)
```

```
## Loading required package: tinytex
```

```
require(dplyr)
```

```
require(tmap)
```

```
## Loading required package: tmap
```

```
require(sf)
```

```
## Loading required package: sf
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
require(cbsodataR)
```

```
## Loading required package: cbsodataR
```

```
require(yaml)
```

```
## Loading required package: yaml
```

Title Page

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Tutorial 1 Group 4

Include your tutorial lecturer's name

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Social Problem: People in Amsterdam who struggle to make ends meet.

Many people living in Amsterdam have difficulty covering their daily expenses, such as rent, food and energy bills. This financial pressure can lead to stress, debt, and a lower quality of life.

Why is this relevant?

This issue is relevant because approximately one in five people in Amsterdam struggles to make ends meet. Rising living costs, high housing prices, inflation, and relatively low incomes are some of the factors contributing to this problem. As a result, many residents face financial insecurity, making it an important social issue that affects both individuals and the wider community.

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

midwest is an example dataset included in the tidyverse package

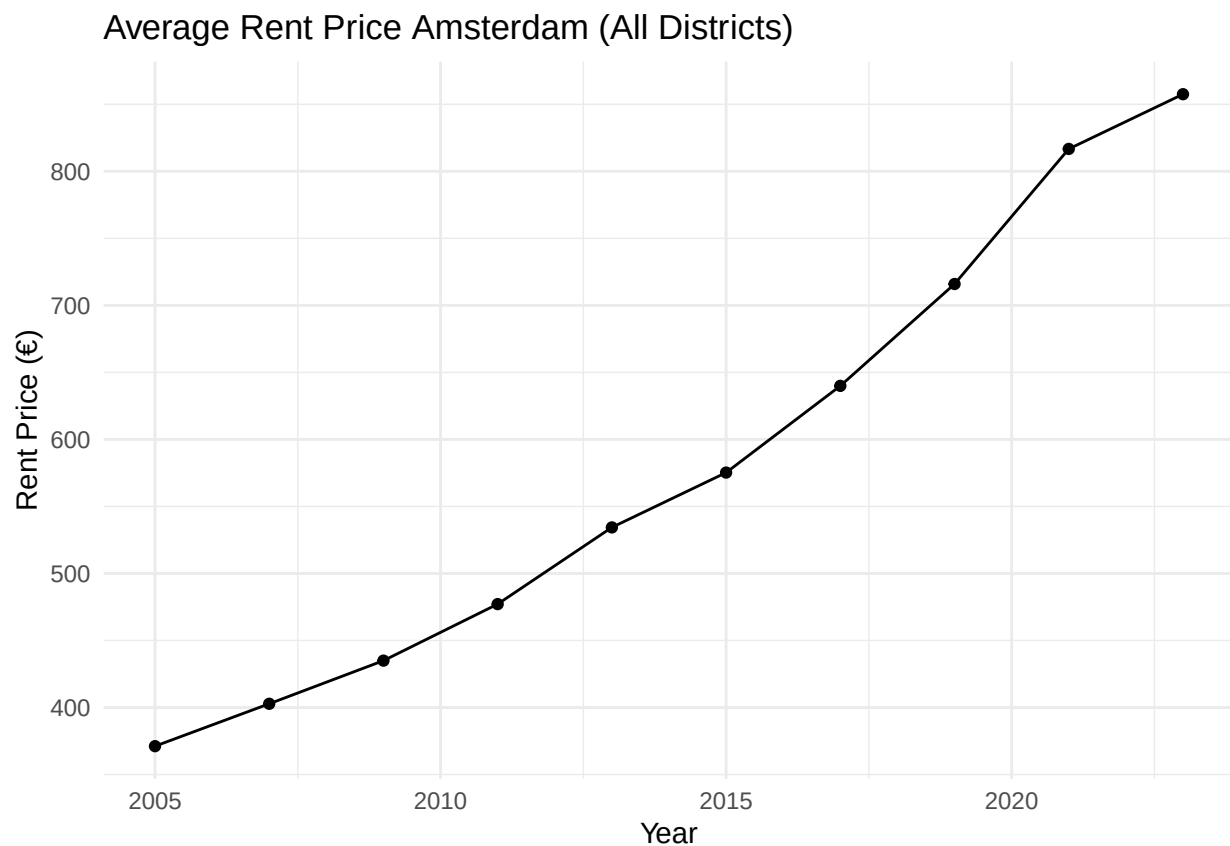
Grafiek 1:

```
library(readxl)
gemiddeldehuur2023 <- read_excel("data/gemiddeldehuur2023.xlsx")
View(gemiddeldehuur2023)

library(dplyr)
library(ggplot2)

per_jaar <- gemiddeldehuur2023 %>%
  group_by(Jaar) %>%
  summarise(gem_huur = mean(`Huur gemiddeld`, na.rm = TRUE))

ggplot(per_jaar, aes(x = Jaar, y = gem_huur)) +
  geom_line() +
  geom_point() +
  labs(
    title = "Average Rent Price Amsterdam (All Districts)",
    x = "Year",
    y = "Rent Price (€)"
  ) +
  theme_minimal()
```



Grafiek 4:

```
moeiteRondkomenAlleJaren <- read.csv2("data/moeiteRondkomenAlleJaren.csv")

moeite_per_jaar <- moeiteRondkomenAlleJaren %>%
```

```
mutate(Jaar = as.numeric(substr(Perioden, 1, 4))) %>%
group_by(Jaar) %>%
summarise(gem_moeite = mean(MoeiteMetRondkomen_32, na.rm = TRUE)) %>%
arrange(Jaar) %>%
mutate(
  procentuele_verandering =
    (gem_moeite - lag(gem_moeite)) / lag(gem_moeite) * 100
)
```

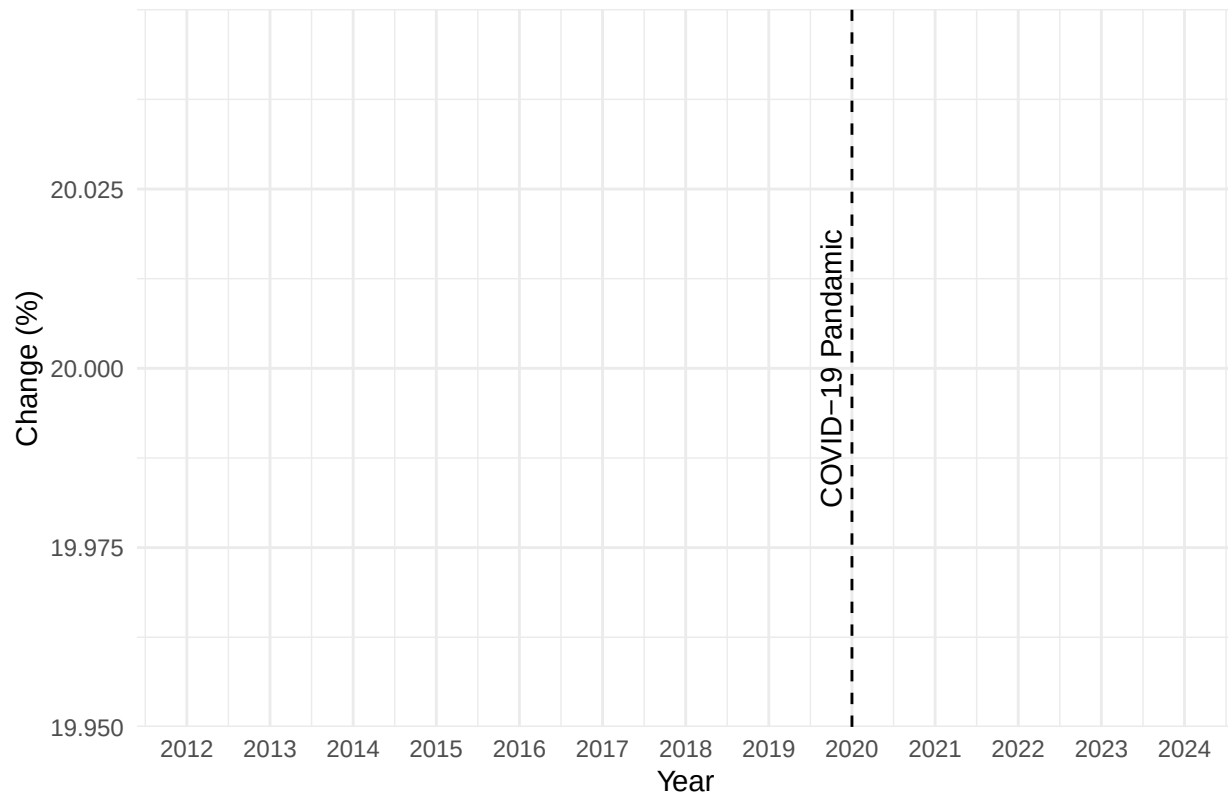
```
## Warning: There were 5 warnings in 'summarise()'.
## The first warning was:
## i In argument: 'gem_moeite = mean(MoeiteMetRondkomen_32, na.rm = TRUE)'.
## i In group 1: 'Jaar = 2012'.
## Caused by warning in 'mean.default()':
## ! argument is not numeric or logical: returning NA
## i Run 'dplyr::last_dplyr_warnings()' to see the 4 remaining warnings.
```

```
ggplot(moeite_per_jaar,
  aes(x = Jaar, y = procentuele_verandering)) +
  geom_line() +
  geom_point() +
  geom_vline(xintercept = 2020, linetype = "dashed") +
  annotate("text",
    x = 2020,
    y = 20,
    label = "COVID-19 Pandemic",
    angle = 90,
    vjust = -0.5) +
  scale_x_continuous(
    breaks = seq(min(moeite_per_jaar$Jaar),
      max(moeite_per_jaar$Jaar),
      by = 1)
  ) +
  labs(
    title = "Percentage Change in Financial Hardship",
    x = "Year",
    y = "Change (%)"
  ) +
  theme_minimal()
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## ('geom_line()').
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Percentage Change in Financial Hardship



```
library(readr)
InkomenWijkenAmsterdam <- read_delim("data/InkomenWijkenAmsterdam.csv",
  delim = ";", escape_double = FALSE, trim_ws = TRUE)

## Rows: 628 Columns: 9
## -- Column specification -----
## Delimiter: ";"
## chr (9): Wijken en buurten, Inkomen/Inkomen van huishoudens/Gem. gestandaard...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

library(readr)
InkomenPerWijk2023Gefilterd <- read_csv("data/InkomenPerWijk2023Gefilterd.csv")

## Rows: 113 Columns: 2
## -- Column specification -----
## Delimiter: ","
## chr (2): Wijken en buurten, Inkomen/Inkomen van huishoudens/Gem. gestandaard...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
View(InkomenPerWijk2023Gefilterd)
```

```
library(readr)
moeiteRondkomenAlleJaren <- read_delim("data/moeiteRondkomenAlleJaren.csv",
  delim = ";", escape_double = FALSE, trim_ws = TRUE)

## Rows: 91550 Columns: 9
## -- Column specification -----
## Delimiter: ";"
## chr (6): Marges, WijkenEnBuurten, Perioden, Gemeentenaam_1, SoortRegio_2, Co...
## dbl (3): ID, Leeftijd, MoeiteMetRondkomen_32
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
library(dplyr)

amsterdam_wijken <- moeiteRondkomenAlleJaren %>%
  filter(Gemeentenaam_1 == "Amsterdam") %>%
  filter(SoortRegio_2 == "Wijk")

amsterdam_wijken_nieuw <- amsterdam_wijken[
  , !(names(amsterdam_wijken) %in% c("ID", "Leeftijd", "Marges"))
]

library(readxl)
Wijken_Amsterdam_2024_CBS <- read_excel("data/Wijken_Amsterdam_2024_CBS.xlsx")
View(Wijken_Amsterdam_2024_CBS)

Wijken_Amsterdam_2024_CBS$Codering_3 <- Wijken_Amsterdam_2024_CBS`CBS Wijkcode`
Wijken_Amsterdam_2024_CBS`CBS Wijkcode` <- NULL

amsterdam_wijken_nieuw2 <- inner_join(Wijken_Amsterdam_2024_CBS, amsterdam_wijken_nieuw, by = "Codering_3")

amsterdam_wijken_nieuw2 <- amsterdam_wijken_nieuw2 %>%
  select(-WijkenEnBuurten, -SoortRegio_2, -Gemeentenaam_1)

View(amsterdam_wijken_nieuw2)
```

```
library(sf)
library(dplyr)
library(ggplot2)
library(readr)
library(cbsodataR)

wijken <- st_read("https://cartomap.github.io/nl/wgs84/wijk_2023.geojson")
```

```
## Reading layer 'wijk_2023' from data source
## 'https://cartomap.github.io/nl/wgs84/wijk_2023.geojson' using driver 'GeoJSON'
## Simple feature collection with 3352 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
```

```
## Bounding box: xmin: 3.358 ymin: 50.751 xmax: 7.227 ymax: 53.554
## Geodetic CRS: WGS 84
```

```
ams <- wijken %>% filter(statcode %in% amsterdam_wijken_nieuw$WijkenEnBuurten)

rondkomen_raw <- read_delim("data/moeiteRondkomenAlleJaren.csv",
  delim = ";",
  locale = locale(decimal_mark = ".")
)
```

```
## Rows: 91550 Columns: 9
## -- Column specification -----
## Delimiter: ";"
## chr (6): Marges, WijkenEnBuurten, Perioden, Gemeentenaam_1, SoortRegio_2, Co...
## dbl (3): ID, Leeftijd, MoeiteMetRondkomen_32
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
rondkomen_2024 <- rondkomen_raw %>%
  filter(
    Perioden == "2024JJ00",
    SoortRegio_2 == "Wijk",
    Gemeentenaam_1 == "Amsterdam"
  ) %>%
  mutate(
    WijkenEnBuurten = trimws(WijkenEnBuurten),
    MoeiteMetRondkomen_32 = as.numeric(MoeiteMetRondkomen_32)
  ) %>%
  select(WijkenEnBuurten, MoeiteMetRondkomen_32)

ams_joined <- ams %>%
  mutate(statcode_clean = trimws(statcode)) %>%
  left_join(rondkomen_2024, by = c("statcode_clean" = "WijkenEnBuurten"))

breaks <- c(6.7, 14.8, 19.8, 23.6, 27.7, 33.4, 36.6)
labels <- c("6,7 - 14,8", "14,8 - 19,8", "19,8 - 23,6",
  "23,6 - 27,7", "27,7 - 33,4", "33,4 - 36,6")

ams_joined <- ams_joined %>%
  mutate(
    rondkomen_cat = cut(
      MoeiteMetRondkomen_32,
      breaks = breaks,
      labels = labels,
      include.lowest = TRUE
    )
  )

kleuren <- c("#fde0d0", "#f7b89c", "#f08060", "#d94f35", "#b81c1c", "#7a0000")

ggplot(ams_joined) +
  geom_sf(aes(fill = rondkomen_cat), color = "white", linewidth = 0.2) +
```

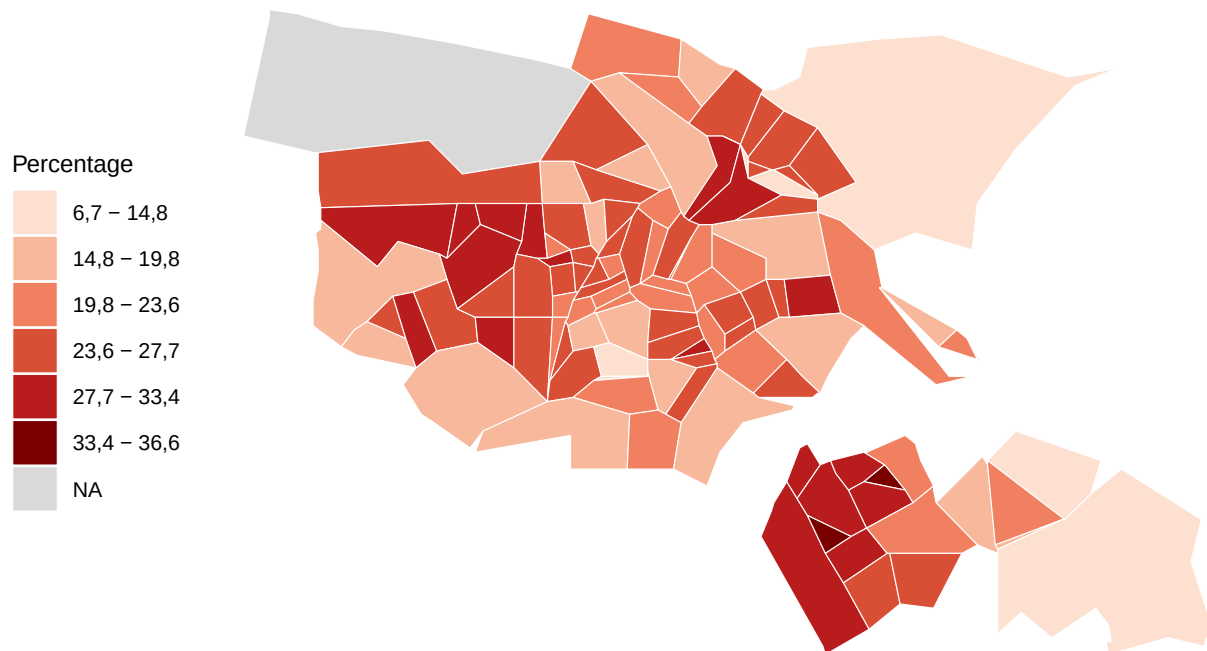
```

scale_fill_manual(
  values = kleuren,
  labels = labels,
  na.value = "grey85",
  name = "Percentage",
  drop = FALSE
) +
theme_void() +
labs(
  title = "Financial Hardship in Amsterdam, 2024",
  subtitle = "Per District, 18+"
) +
theme(
  plot.title = element_text(face = "bold", size = 13, color = "#1a4a8a"),
  plot.subtitle = element_text(size = 9, color = "#444444"),
  legend.title = element_text(size = 9),
  legend.text = element_text(size = 8),
  legend.position = "left"
)

```

Financial Hardship in Amsterdam, 2024

Per District, 18+



```

library(sf)
library(dplyr)
library(ggplot2)
library(readr)
library(cbsodataR)

```



```

wijken <- st_read("https://cartomap.github.io/nl/wgs84/wijk_2023.geojson")

## Reading layer 'wijk_2023' from data source
## 'https://cartomap.github.io/nl/wgs84/wijk_2023.geojson' using driver 'GeoJSON'
## Simple feature collection with 3352 features and 6 fields
## Geometry type: MULTIPOLYGON
## Dimension: XY
## Bounding box: xmin: 3.358 ymin: 50.751 xmax: 7.227 ymax: 53.554
## Geodetic CRS: WGS 84

ams <- wijken %>% filter(grepl("^WK0363", statcode))

leeftijd_raw <- read_delim("data/Leefijdsgroepen_wijken_buurtten_Amsterdam_2023.csv",
  delim = ";",
  locale = locale(decimal_mark = ".")
)

## Rows: 633 Columns: 7
## -- Column specification -----
## Delimiter: ";"
## chr (1): Wijken en buurten
## dbl (6): Bevolking/Aantal inwoners (aantal), Bevolking/Leefijdsgroepen/0 to...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

leeftijd_wijk <- leeftijd_raw %>%
  rename(
    wijknaam = `Wijken en buurten`,
    inwoners = `Bevolking/Aantal inwoners (aantal)`,
    leeftijd_15_25 = `Bevolking/Leefijdsgroepen/15 tot 25 jaar (aantal)`
  ) %>%
  mutate(wijknaam = trimws(wijknaam)) %>%
  distinct(wijknaam, .keep_all = TRUE) %>%
  filter(inwoners > 0) %>%
  mutate(perc_studenten = leeftijd_15_25 / inwoners * 100) %>%
  select(wijknaam, perc_studenten)

ams_joined <- ams %>%
  left_join(leeftijd_wijk, by = c("statnaam" = "wijknaam"))

breaks <- c(0, 8, 11, 13, 16, 20, 75)
labels <- c("0 - 8", "8 - 11", "11 - 13", "13 - 16", "16 - 20", "20+")

ams_joined <- ams_joined %>%
  mutate(
    studenten_cat = cut(
      perc_studenten,
      breaks = breaks,
      labels = labels,
      include.lowest = TRUE
    )
  )

```

```

    )
  )

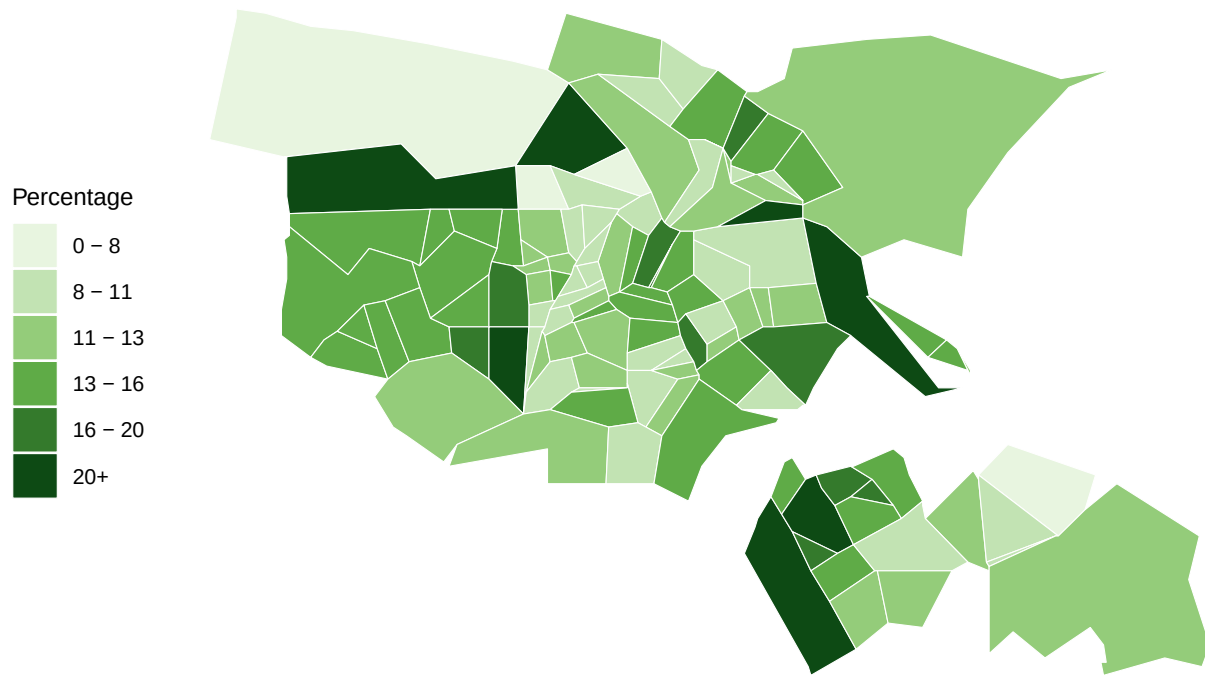
kleuren <- c("#e8f5e0", "#c2e3b3", "#94cc7a", "#5fab47", "#347a2c", "#0d4a14")

ggplot(ams_joined) +
  geom_sf(aes(fill = studenten_cat), color = "white", linewidth = 0.2) +
  scale_fill_manual(
    values = kleuren,
    labels = labels,
    na.value = "grey85",
    name = "Percentage",
    drop = FALSE
  ) +
  theme_void() +
  labs(
    title = "young adults/students in Amsterdam, 2023",
    subtitle = "Percentage 15-25 years old per district"
  ) +
  theme(
    plot.title = element_text(face = "bold", size = 13, color = "#1a4a8a"),
    plot.subtitle = element_text(size = 9, color = "#444444"),
    legend.title = element_text(size = 9),
    legend.text = element_text(size = 8),
    legend.position = "left"
  )

```

young adults/students in Amsterdam, 2023

Percentage 15–25 years old per district



2.2 Provide a short summary of the dataset(s)

2.3 Describe the type of variables included

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

3.2 Generate necessary variables

Variable 1

Variable 2

3.3 Visualize temporal variation

3.4 Visualize spatial variation

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

The findings indicate that financial hardship in Amsterdam is closely related to rising living costs and differences in income across neighbourhoods. The rental price data show a strong increase in average rents between 2005 and 2023, with average rents more than doubling during this period. This suggests that housing has become significantly less affordable, especially for households whose incomes have not increased at the same pace.

The income dataset reveals substantial differences in average household income between Amsterdam neighbourhoods. Some neighbourhoods have considerably higher average incomes than others, indicating that residents do not experience rising living costs in the same way. For households in lower-income neighbourhoods, increasing housing costs are likely to have a greater impact on their financial situation because a larger share of their income must be spent on essential expenses.

The dataset on difficulties making ends meet provides further evidence that financial vulnerability remains an important issue in Amsterdam. When viewed together, the income, housing, and financial hardship data suggest that residents with lower incomes are more likely to experience financial pressure. Rising rents, combined with increasing costs for food, energy, and other necessities, may leave these households with limited financial flexibility.

External factors should also be considered. The COVID-19 pandemic affected employment, income security, and economic activity, while the subsequent period of inflation increased the cost of living. These developments likely intensified financial difficulties for many residents and may help explain why a significant share of Amsterdam's population continues to struggle to make ends meet.

A limitation of this analysis is that it does not include all possible determinants of financial wellbeing, such as debt levels, savings, household composition, or educational attainment. Nevertheless, the combined evidence from the datasets suggests that rising housing costs and unequal income levels are key factors contributing to financial insecurity in Amsterdam.

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.